

Development Team Project: Report

Introduction

Airbnb is an online accommodation marketplace platform that connects millions of tourists with hosts providing entire houses, apartments, and rooms to them, offering an alternative accommodation option to traditional hotels. The site had over 275 million users and recorded over 490 million booked nights in 2024 alone, making it a leading player in the travel industry (Piva, 2025).

Understanding which characteristics influence a listing's price is a key activity for Airbnb's executive team, as these insights directly inform decisions and activities around which customer segments and geographical markets to direct marketing materials towards. It can also support targeted outreach to property owners, helping to identify hosts with certain unit types in particular locations which are most likely to be booked and result in revenue for the platform.

Machine learning (ML) models can be used to determine this. This report details the analysis conducted on Airbnb's dataset, focused on listings within the New York Area in the United States of America in 2019 (*New York City Airbnb Open Data*, no date), and documents the subsequent build of machine learning models to answer the following research question: which factors influence the revenue potential of a listing?

Exploratory Data Analysis and Data Preprocessing

Developing machine learning models from messy data containing missing values and outliers would reduce the model's accuracy and would provide Airbnb's team with inaccurate predictions

and misleading insights on the factors influencing a listing's price (Côté *et al.*, 2024). Initial exploration of the dataset using the pandas function `'describe()'` confirmed the presence of 16 attributes and 48,995 listings. Summary statistics for the 10 numeric variables were retrieved at this point, including the mean values of listing price (\$152.72), minimum number of nights (7), and the availability of the listing per year (112.78). The number of missing values within each variable were also discovered with the majority of them located in the review-related attributes.

To clean the data, missing values were filled, with values in the `'reviews_per_month'` variable imputed with 0. Price outliers located outside of the 5th and 90th percentiles were dropped to avoid inflating error metrics.

Further exploratory data analysis (EDA) enabled pattern identification between the variables to inform the development of the specific model. This is another important step to prevent issues such as multicollinearity from affecting the final models. The geospatial plot in figure 1 shows that most of the listings were in Manhattan, with fewer listings in the Bronx and Staten Island. Additionally, bivariate analyses were conducted to visualise the relationship between variables. Distribution of the listing price by `'room_type'` showed that entire homes or apartments had a median price of \$150 per night, followed by private rooms and shared rooms at \$74 and \$48 respectively (figure 2). The price by `'neighbourhood_group'` shown in figure 3 confirms that Manhattan is the most expensive area with listings surpassing \$140 per night on average. Finally, a correlation matrix (figure 4) showed overall weak positive & weak negative correlations for all variables, except for `'number_of_reviews'` and `'reviews_per_month'`, reporting a coefficient of 0.59 indicating a moderate positive one.

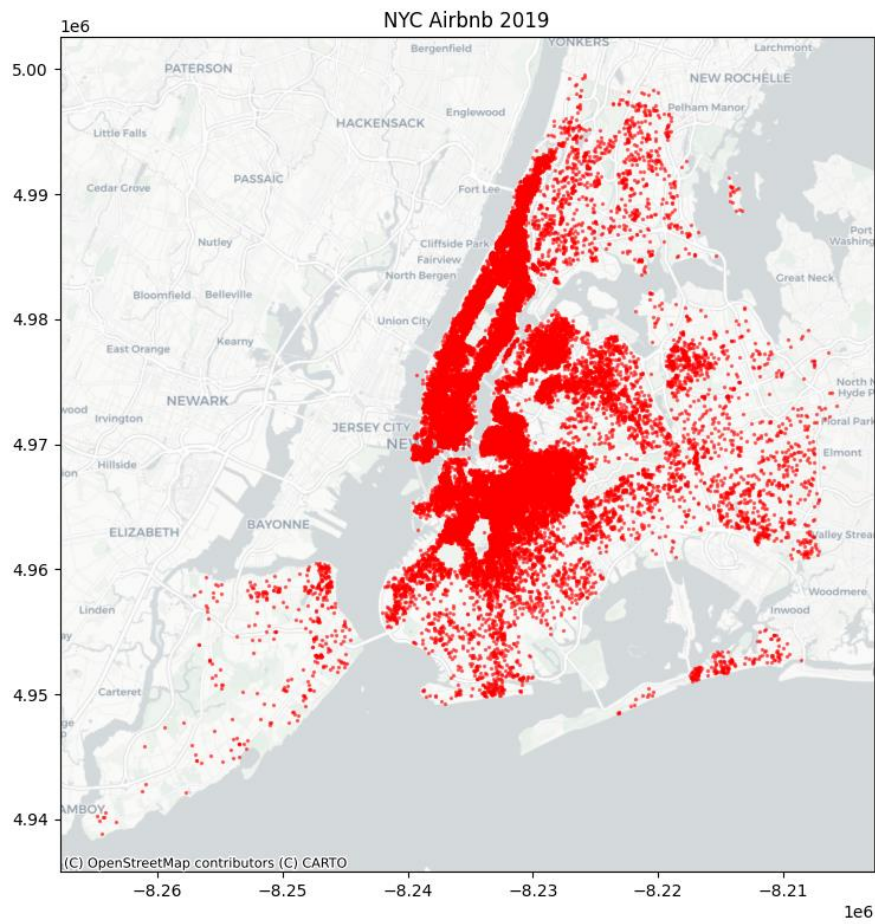


Figure 1 - geospatial plot of all listings

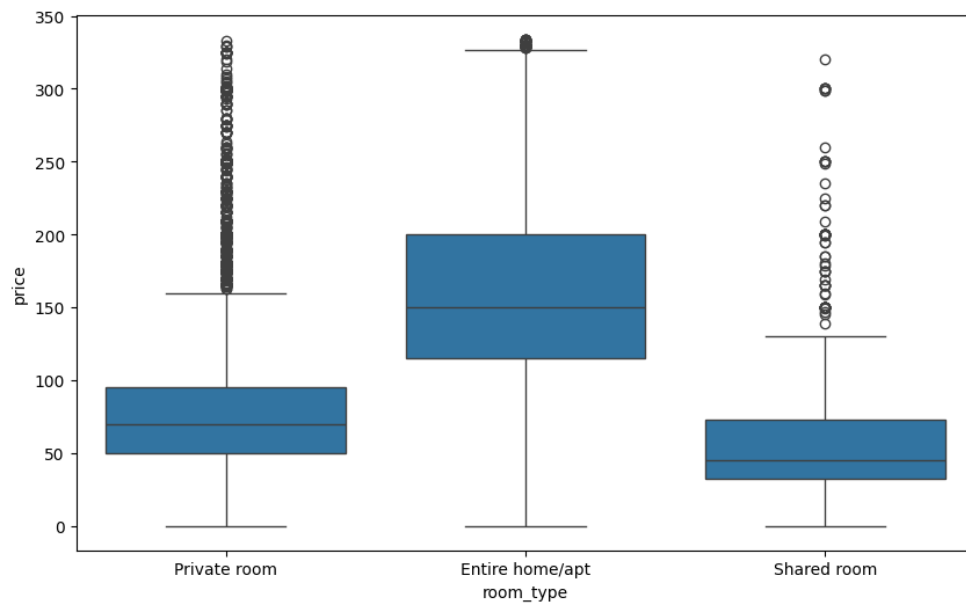


Figure 2 - box plots displaying the distribution of price by room type

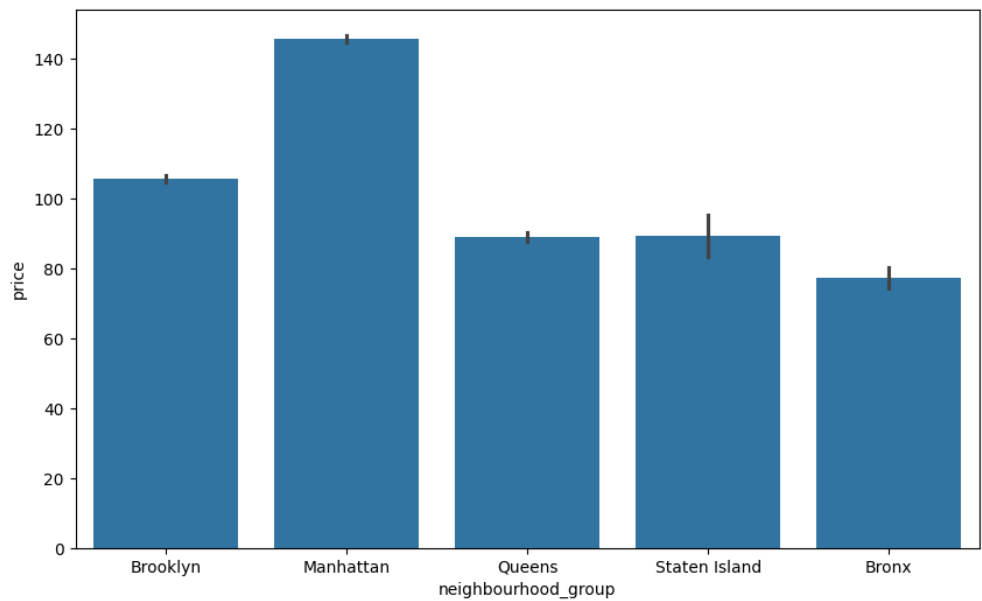


Figure 3 - bar plot displaying mean price by neighbourhood group

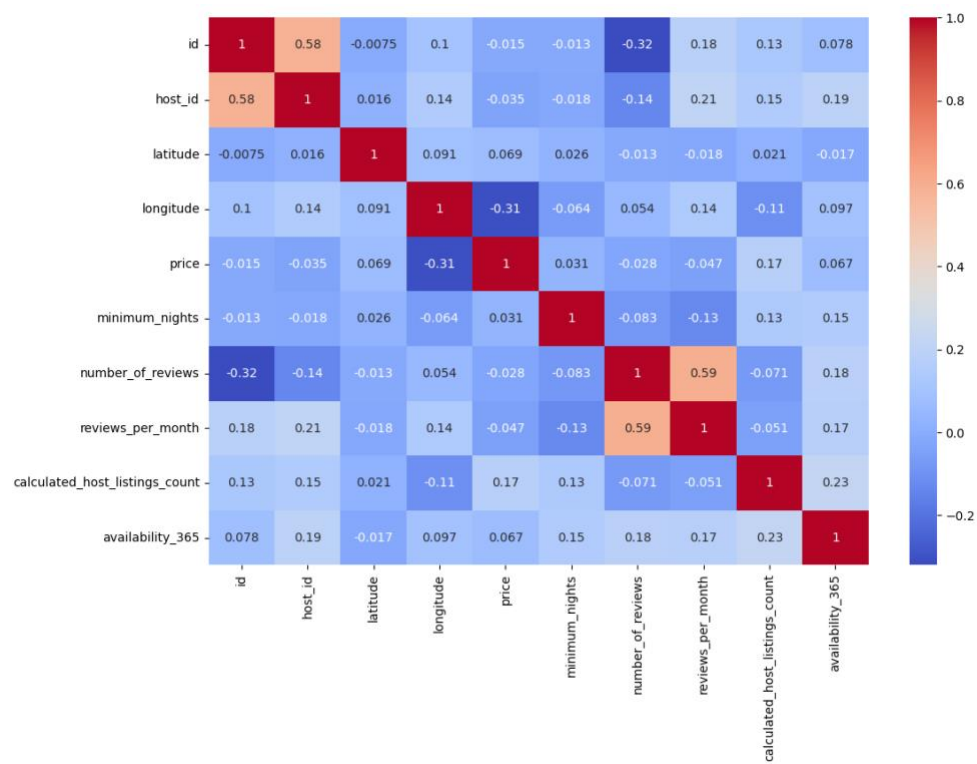


Figure 4 - correlation heatmap of numeric variables

The final step before model development was data preprocessing. As ML models cannot process numerical variables, categorical variables such as 'room_type' were one-hot encoded to convert them into binary columns. Features selected were standardised using 'StandardScaler' to reduce the chance of inaccuracy and reliability in the final model predictions (Ahsan *et al.*, 2021).

Model Development and Evaluation

For the regression analysis, the target variable was listing price. Predictor variables included room type, neighbourhood group, minimum nights, reviews per month, availability_365, and calculated host listings count. The model identified strong relationships between location and pricing, with Manhattan accounting for 21,661 listings (44.3%) and exhibiting significantly higher average prices than other boroughs. Brooklyn represented 20,104 listings (41.1%), while Queens accounted for 5,666 listings (11.6%). These geographical differences contributed substantially to pricing variance within the dataset.

The clustering phase employed K-Means segmentation to categorise Airbnb properties into distinct market groups. Feature selection focused on pricing, availability, review frequency, and minimum stay requirements. The optimal number of clusters was determined through the Elbow Method, producing clear segmentation between premium, mid-range, and budget-oriented listings. Cluster analysis revealed that premium properties were characterised by high prices, limited availability, and strategic locations within Manhattan, whereas budget clusters were concentrated in outer boroughs with higher availability and lower occupancy costs.

Model evaluation demonstrated satisfactory predictive performance. Regression results indicated that room type, neighbourhood group, and availability were the most influential predictors of listing price. Feature importance analysis showed that geographical location contributed the highest explanatory power, confirming that market value is strongly influenced by borough-level demand patterns. The clustering model achieved well-separated customer segments, providing meaningful business insights for market targeting and pricing optimisation (figure 5).



Figure 5 - the four clusters visualised using principal component analysis

From a business analytics perspective, the models generated actionable intelligence for Airbnb. Regression outputs support dynamic pricing decisions by estimating expected listing values, while clustering results enable customer segmentation strategies that can improve marketing effectiveness and host recommendations. The combined machine learning approach therefore provides both predictive and descriptive analytics capabilities, supporting data-driven decision-making within the Airbnb marketplace.

Conclusion

This project successfully applied Classical ML techniques, specifically regression and clustering, to analyse Airbnb business trends within the New York City market. Using a dataset of 48,895 listings, the study identified significant pricing determinants and generated meaningful customer segments through unsupervised learning techniques. Regression analysis demonstrated that variables such as 'neighbourhood_group', room type, and property availability have a substantial impact on listing prices, while clustering analysis revealed distinct premium, mid-market, and budget property categories.

The results highlight the practical value of machine learning in supporting Airbnb's strategic objectives. Predictive modelling can improve pricing accuracy, increase revenue optimisation opportunities, and assist hosts in setting competitive rates. Furthermore, clustering enables more targeted marketing campaigns and enhanced customer personalisation by identifying behavioural and market-based segments. Overall, the developed models demonstrate how machine learning can transform large-scale accommodation data into actionable business intelligence, providing Airbnb executives with evidence-based insights for operational and strategic decision-making.

References

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